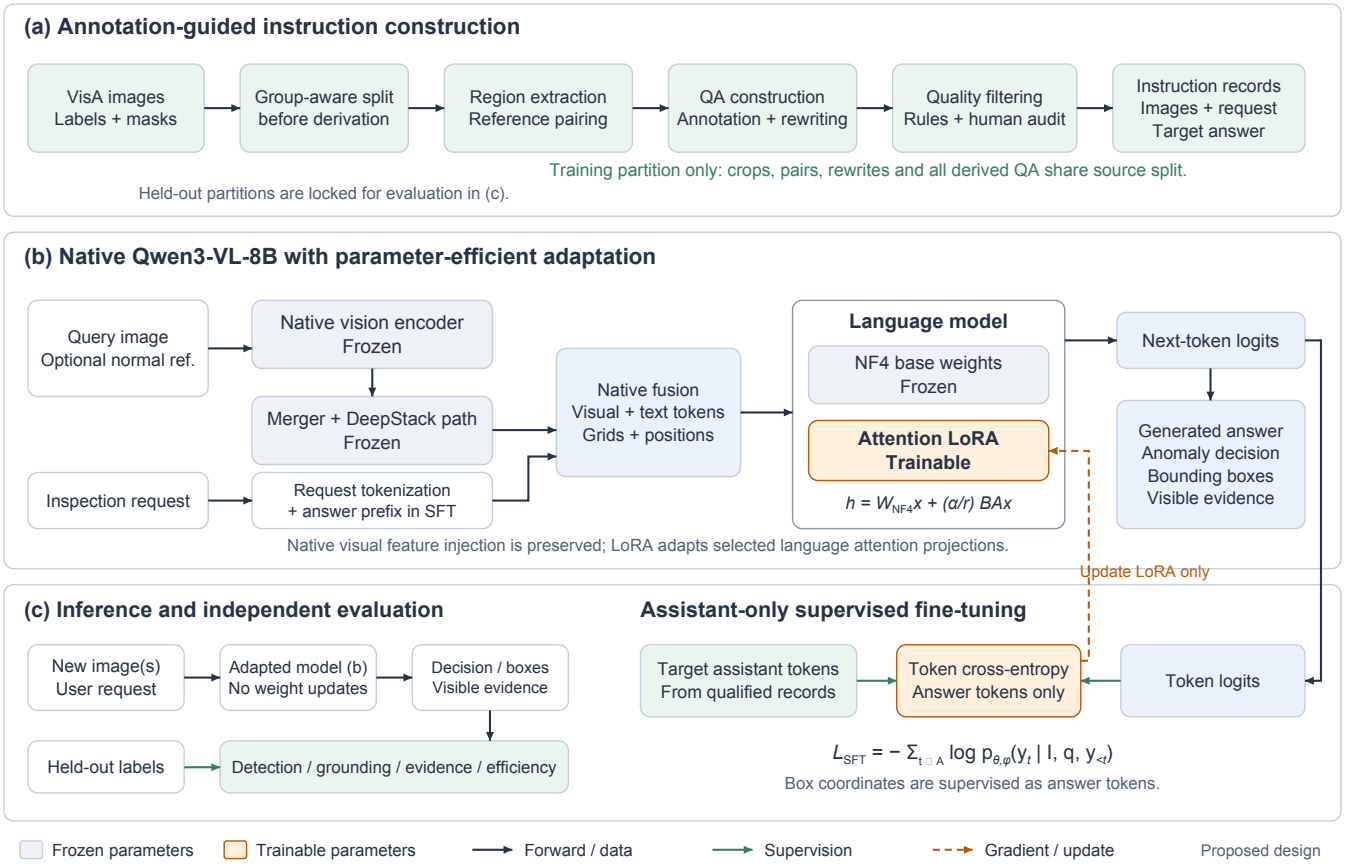

(a) Annotation-guided instruction construction
VisA images
Labels + masks
Group-aware split
before derivation
Region extraction
Reference pairing
QA construction
Annotation + rewriting
Quality filtering
Rules + human audit
Instruction records
Images + request
Target answer
Training partition only: crops, pairs, rewrites and all derived QA share source split.
Held-out partitions are locked for evaluation in (c).
(b) Native Qwen3-VL-8B with parameter-efficient adaptation
Query image
Optional normal ref.
Native vision encoder
Frozen
Merger + DeepStack path
Frozen
Inspection request
Request tokenization
+ answer prefix in SFT
Native fusion
Visual + text tokens
Grids + positions
Language model
NF4 base weights
Frozen
Attention LoRA
Trainable
h = W_NF4x + (α/r) BAx
Next-token logits
Generated answer
Anomaly decision
Bounding boxes
Visible evidence
Native visual feature injection is preserved; LoRA adapts selected language attention projections.
Update LoRA only
(c) Inference and independent evaluation
New image(s)
User request
Adapted model (b)
No weight updates
Decision / boxes
Visible evidence
Held-out labels
Detection / grounding / evidence / efficiency
Assistant-only supervised fine-tuning
Target assistant tokens
From qualified records
Token cross-entropy
Answer tokens only
Token logits
L_SFT = − Σ_{t ∈ A} log p_{θ,φ}(y_t | I, q, y_<t)
Box coordinates are supervised as answer tokens.
Frozen parameters
Trainable parameters
Forward / data
Supervision
Gradient / update
Proposed design